GENERAL DESCRIPTION

The adc1395x_pc is a CMOS 3.3V 10-bit analog-to-digital converter (ADC). It converts the analog input signal into 10-bit binary digital codes at a maximum conversion rate of 500KSPS with 2.5MHz clock.

The device is a recycling type monolithic ADC with an on-chip sample-and-hold function. The ADC has power down mode.

FEATURES

— Resolution: 10-bit

— Maximum conversion rate: 500KSPS

— Main clock: 2.5MHz (max.)

— Power supply: 3.3V ± 0.3V

— Total current: 20uA (Standby mode)
2.0mA (Normal operation)

— Input range: 0.0V ~ 3.3V (3.3VP-P)

— Differential linearity error: ± 1.0 LSB

— Integral linearity error: ± 2.0 LSB

— Signal to noise & distortion ratio: 54dB (Typ.)

— Digital output: CMOS Level

— Operating temperature range: -40°C ~ 85°C

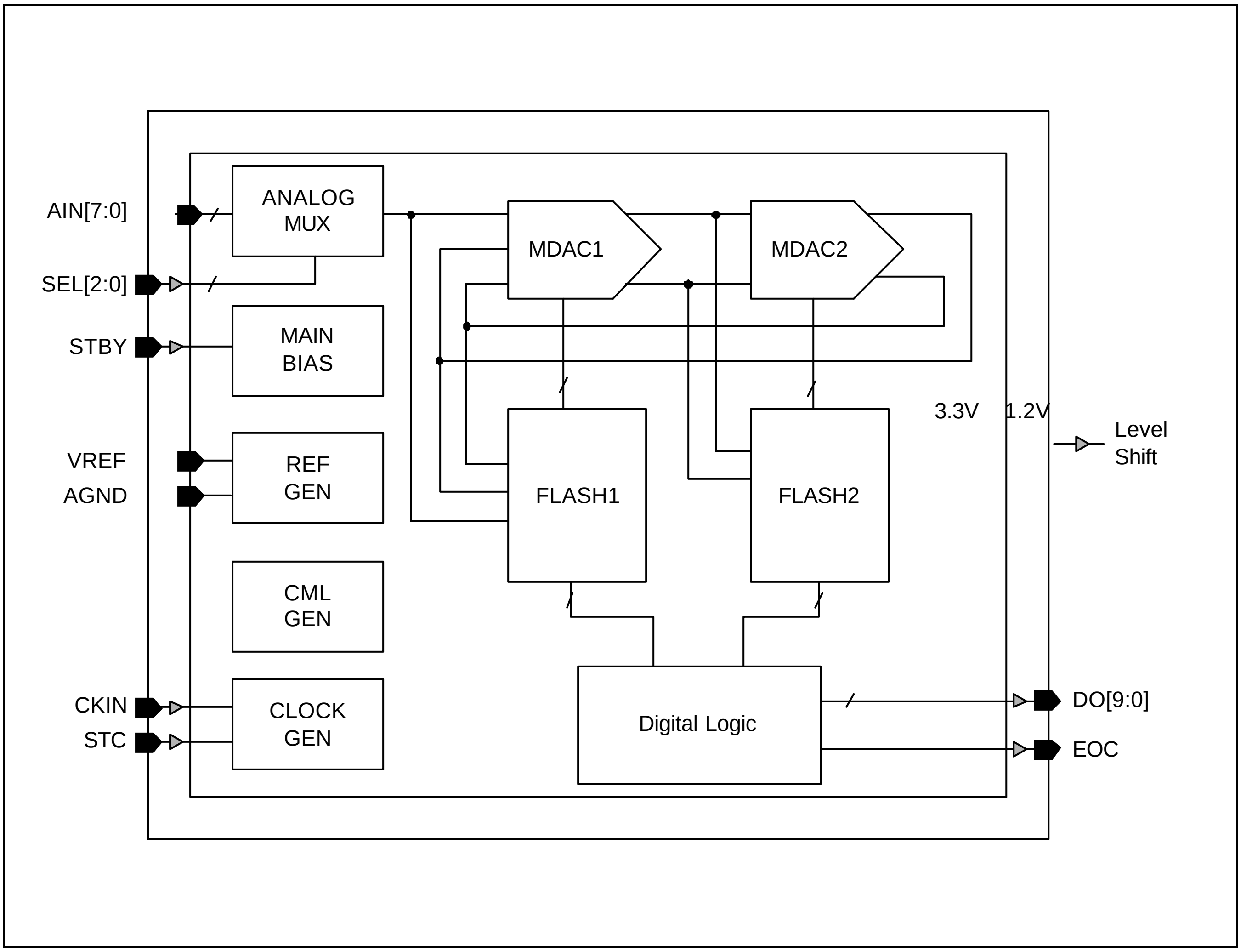
TYPICAL APPLICATIONS

— MICOM interface

— Portable equipment

— Low-power application

FUNCTIONAL BLOCK DIAGRAM



Ver 1.2 (Feb. 2004)

CORE PIN DESCRIPTION

Name	I/O Type	I/O Pad	Pin Description
VREF	AI	phia_abb	Reference top (3.3V)
AGND	AI	phia_abb	Reference bottom (0.0V)
AVDD33A1	AP	vdd33th_abb	Analog power (3.3V)
AVBB33A1	AG	vbbh_abb	Analog sub bias (0.0V)
AVSS33A1	AG	vssth_abb	Analog ground (0.0V)
AIN[7:0]	AI	phiar50_abb	Analog input (input range: 0.0V ~ 3.3V)
SEL[2:0]	DI	internal	Analog input selection pin
STBY	DI	internal	VDD = Power saving (standby), GND = Normal
CKIN	DI	internal	Sampling clock input
DO[9:0]	DO	internal	Digital output
EOC	DO	internal	End of conversion signal
STC	DI	internal	Start of conversion signal
AVSS33A2	DG	vssth_abb	Digital ground (0.0V)
AVDD33A2	DP	vdd33th_abb	Digital power (3.3V)
AVDD12	DP	internal	Digital power (1.2V)

I/O Type Abbr.

— AI: Analog Input

— DI: Digital Input

— AO: Analog Output

— DO: Analog Output

— AP: Analog Power

— AG: Analog Ground

— DP: Digital Power

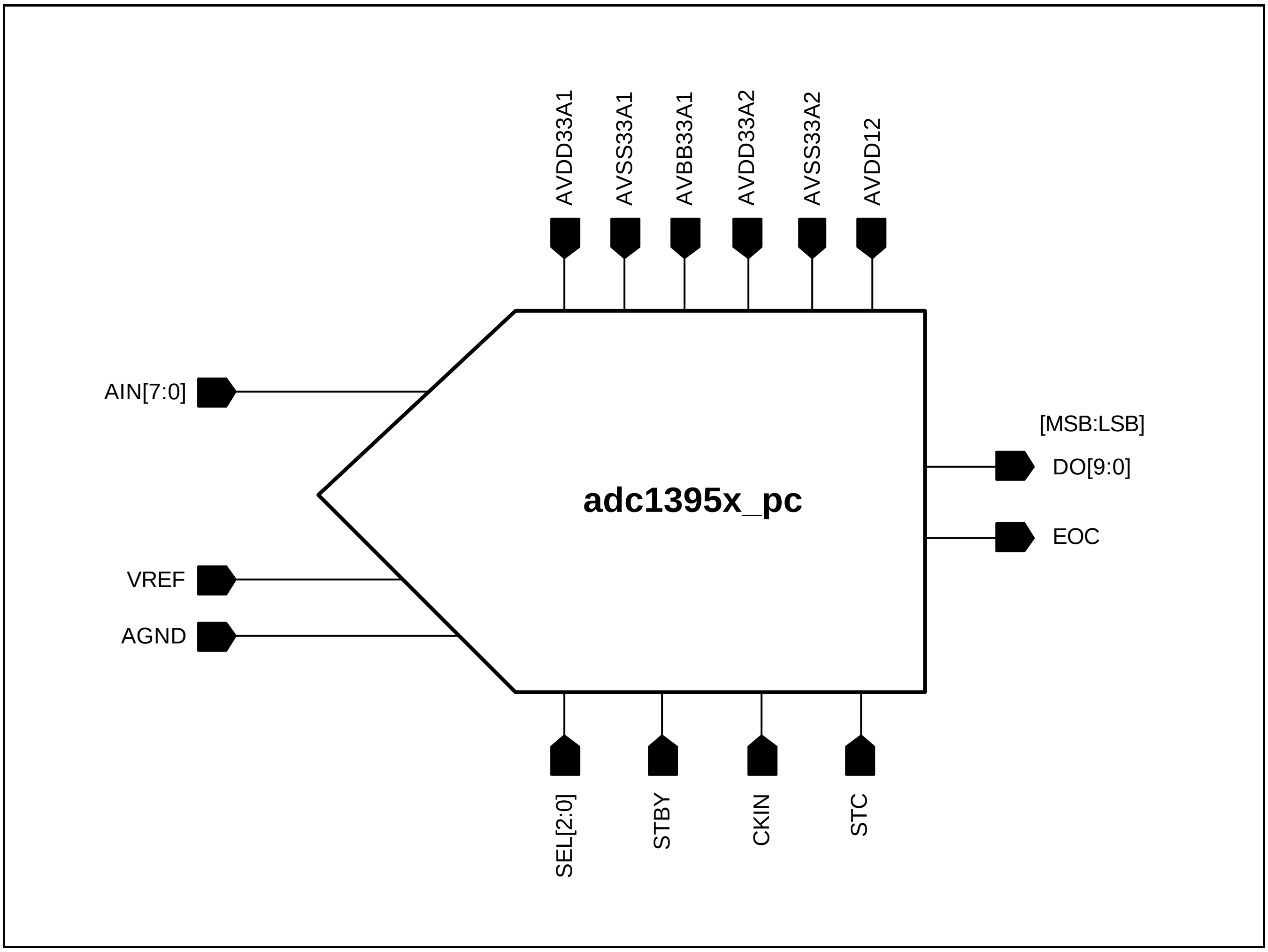
— DG: Digital Ground

— AB: Analog Bi-Direction

— DB: Digital Bi-Direction

NOTE: "internal" means that the usage of the core pins is restricted to the internal connection with the related other blocks. To use the external pads, consult with SEC about the additional issues.

CORE CONFIGURATION



ABSOLUTE MAXIMUM RATINGS

Characteristics	Symbol	Value	Unit
Supply voltage	VDD33	4.8	V
Analog input voltage	AINT	VSS to VDD	V
Digital input voltage	CKIN	VSS to VDD	V
Reference voltage	VREF / AGND	VSS to VDD	V
Storage temperature range	Tstg	-45 to 150	°C
Operating temperature range	Topr	-40 to 85	°C

NOTES:
1. Absolute maximum rating specifies the values beyond which the device may be damaged permanently. Exposure to absolute maximum rating conditions for extended periods may affect reliability. Each condition value is applied with the other values kept within the following operating conditions and function operation under any of these conditions is not implied.
2. All voltages are measured with respect to VSS unless otherwise specified.
3. 100pF capacitor is discharged through a 1.5kΩ resistor (Human body model)

RECOMMENDED OPERATING CONDITIONS

Characteristics	Symbol	Min	Typ	Max	Unit
Supply voltage	AVDD33A1 AVDD33A2	3.0	3.3	3.6	V
Reference input voltage	VREF AGND	2.0 0.0	3.3 0.0	3.6 0.0	V
Analog input voltage	AINT	0.0	VREF	-	V
Operating temperature	Toper	-40	-	85	°C

NOTE: It is strongly recommended that all the supply pins (AVDD33A1, AVDD33A2) be powered from the same source to avoid power latch-up.

DC ELECTRICAL CHARACTERISTICS

Characteristics	Symbol	Min	Typ	Max	Unit	Test Condition
Differential nonlinearity	DNL	-	± 0.8	± 1	LSB	VREF = 3.3V AGND = 0.0V
Integral nonlinearity	INL	-	± 1.0	± 2	LSB	VREF = 3.3V AGND = 0.0V
Offset voltage	TOPOFF BOTOFF	-	3	8	LSB	VREF = 3.3V AGND = 0.0V

(Converter Specifications: AVDD33A1 = AVDD33A2 = 3.3V, AVDD12 = 1.2V, AVSS33A1 = AVSS33A2 = 0V, Toper = 25°C, VREF = 3.3V, AGND = 0.0V unless otherwise specified)

AC ELECTRICAL CHARACTERISTICS

Characteristics	Symbol	Min	Typ	Max	Unit	Test Condition
Maximum conversion rate	fc	-	-	500	KSPS	f_{CKIN} = 2.5MHz
Standby supply current	-	-	20	60	uA	STBY = VDD
Dynamic supply current	IVDD	-	2.0	3	mA	f_{CKIN} = 2.5MHz (without system load)
Reference current	IREF	-	0.4	0.6	mA	V_{REF} = 3.3V
Total harmonic distortion	THD	-	-60	-56	dB	f_{CKIN} = 2.5MHz AINT = 100kHz
Signal-to-noise & distortion ratio	SNDR	48	54	-	dB	f_{CKIN} = 2.5MHz AINT = 100kHz

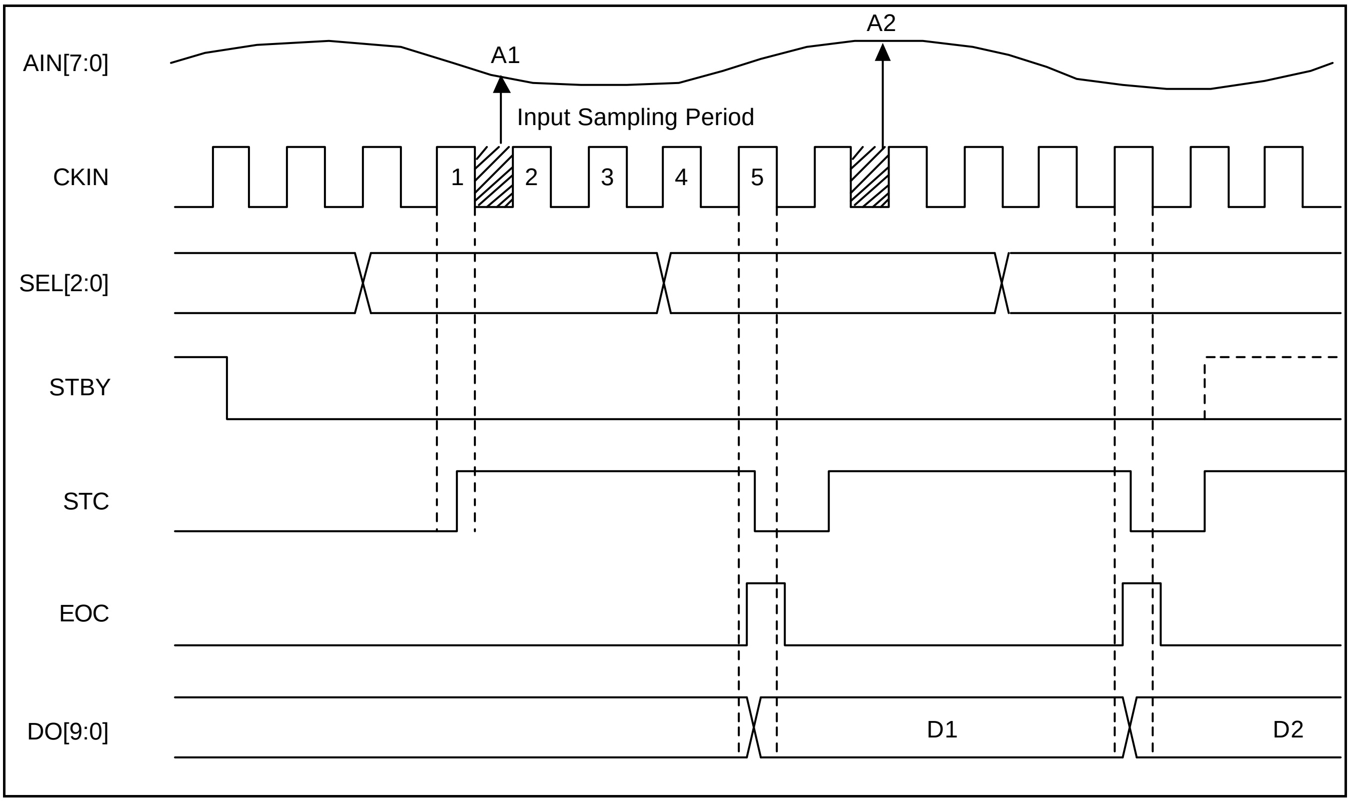
(Converter Specifications : AVDD33A1 = AVDD33A2 = 3.3V, AVDD12 = 1.2V, AVSS33A1 = AVSS33A2 = 0V, Toper = 25°C, VREF = 3.3V, AGND = 0.0V unless otherwise specified)

I/O CHART

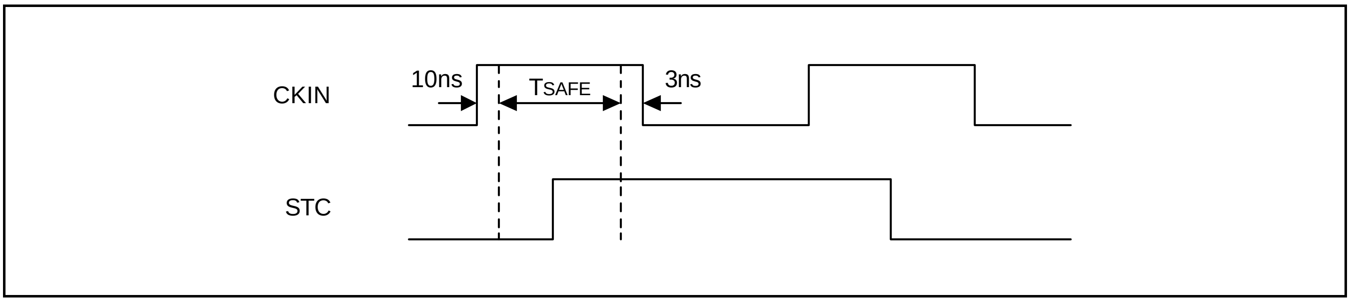
Index	AINT Input (V)	Digital Output	
0	~ 0.00322	00 0000 0000	1LSB = 3.22mV
1	0.00322 ~ 0.00644	00 0000 0001	VREF = 3.3V
2	0.00644 ~ 0.00967	00 0000 0010	AGND = 0.0V
~	~	~	
511	1.64678 ~ 1.65000	01 1111 1111	
512	1.65000 ~ 1.65322	10 0000 0000	
513	1.65322 ~ 1.65644	10 0000 0001	
~	~	~	
1021	3.29033 ~ 3.29355	11 1111 1101	
1022	3.29355 ~ 3.29678	11 1111 1110	
1023	3.29678 ~	11 1111 1111	

TIMING DIAGRAM

1. Main Waveform

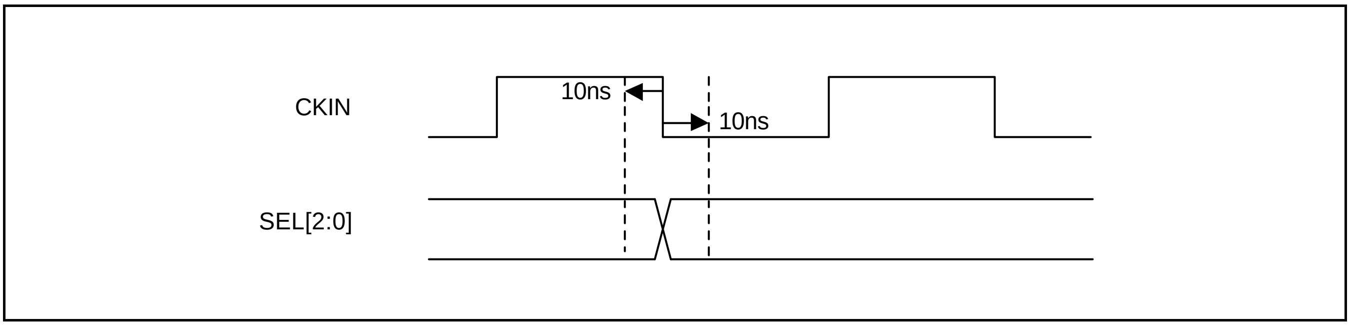


2. STC & CKIN Condition



The A/D Converter operates data conversion when STC (Start Conversion) signal is just "HIGH". Otherwise, output data (DO[9:0]) keep the current states. The STC signal should be changed during "T_{SAFE}" with the "HIGH" level of the clock to operation as shown in the main waveform.

3. SEL[2:0] & CKIN Condition



The transition of SEL[2:0] signals should be synchronized with the falling edge of CKIN signal.

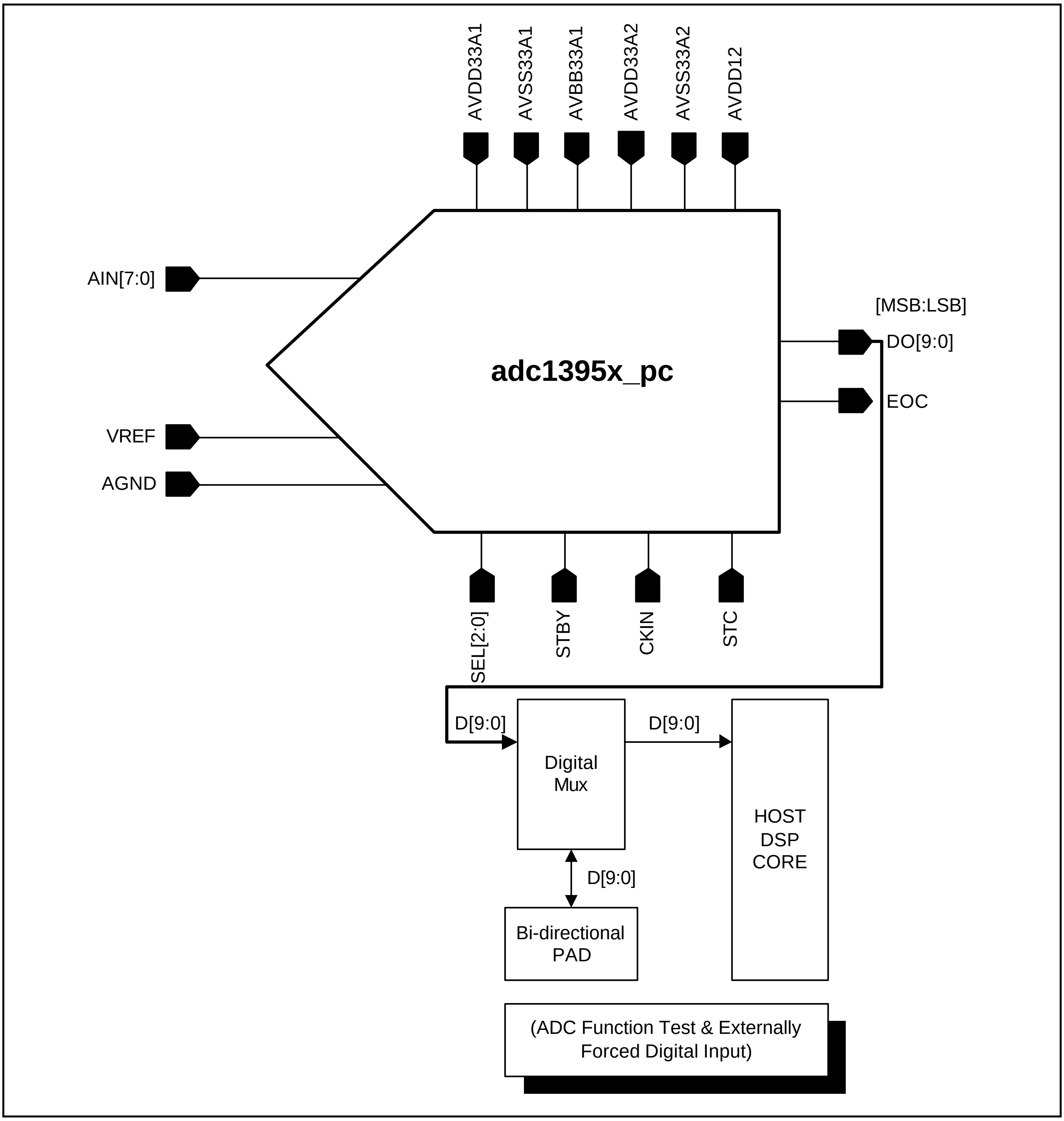
This ADC samples analog input signal at the first low state of CKIN after rising transition of STC.

So you should keep in mind that a proper channel should be selected by SEL[2:0] before sampling the analog input signal.

CORE EVALUATION GUIDE

ADC function is evaluated by external check on the bi-directional pads connected to input nodes of HOST DSP back-end circuit.

The reference voltages may be biased internally through resistor divider.



USER GUIDE

Input Range

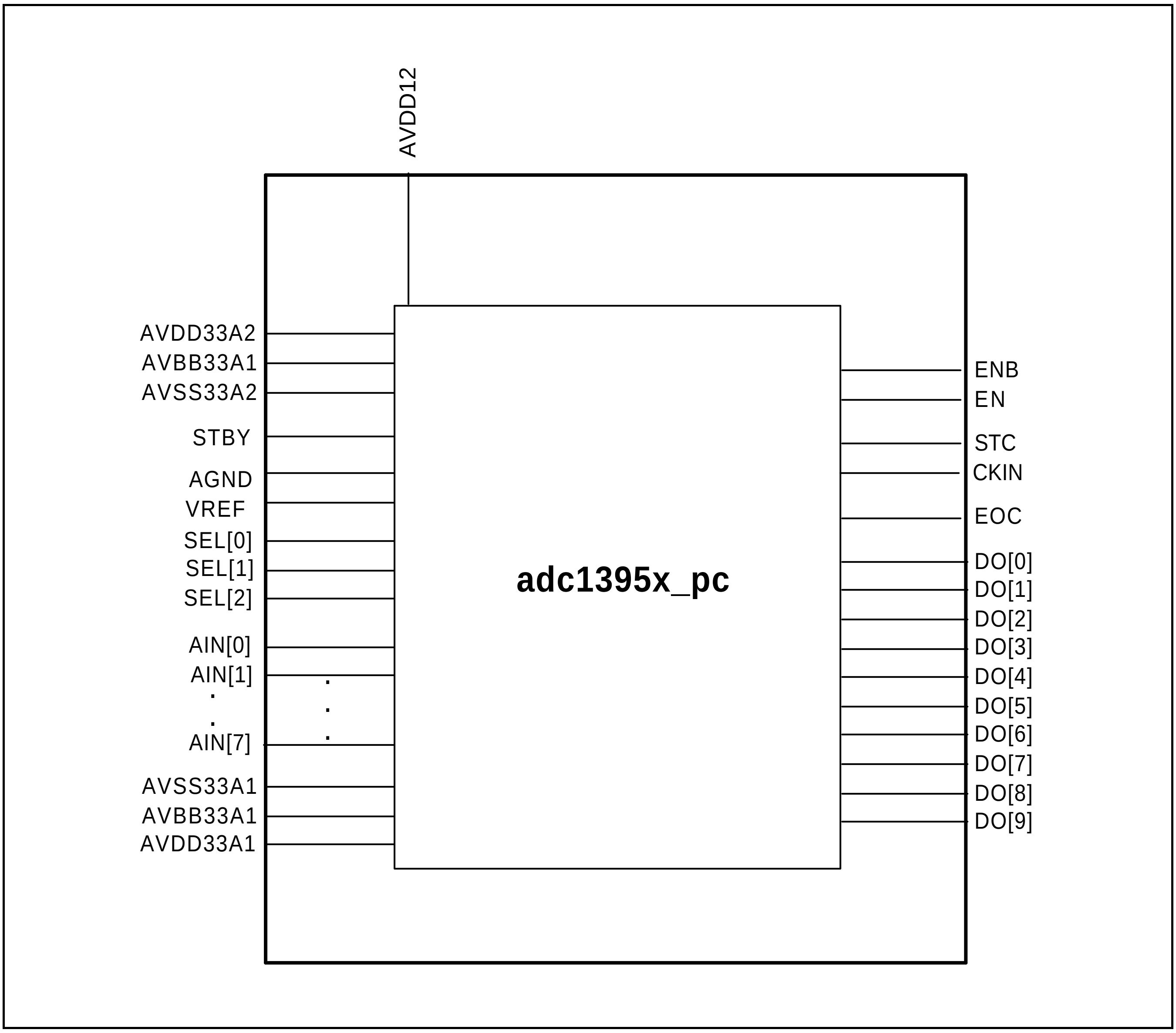
The analog input is single-ended type and the range is from VREF to AGND. This AINT voltage follows reference voltage range fundamentally. So, if you want to alter into the another input range, you should change the voltage value of VREF.

You can use the AINT voltage whose minimum range is 2.0V. In this case, the VREF is 2.0V.

ANALOG INPUT SELECTION TABLE

	AIN[7]	AIN[6]	AIN[5]	AIN[4]	AIN[3]	AIN[2]	AIN[1]	AIN[0]
SEL[2]	1	1	1	1	0	0	0	0
SEL[1]	1	1	0	0	1	1	0	0
SEL[0]	1	0	1	0	1	0	1	0

PHANTOM CELL INFORMATION



Name	I/O Type	Pin Usage	Pin Description
AIN[7:0]	AI	Internal / External	AIN[7:0] signal should not be crossed by any signals and should not run next to digital signals to minimize capacitive coupling between the two signals.
SEL[2:0]	DI	Internal / External	Digital Input Signal lines must have same length to reduce propagation delay.
STBY	DI	Internal / External	
CKIN	DI	Internal / External	
DO[9:0]	DO	Internal / External	
EOC	DO	Internal / External	
STC	DI	Internal / External	
VREF	AI	External	Voltage reference lines (VREF and AGND) must be wide metal to reduce voltage drop of metal lines.
AGND	AI	External	
AVDD33A1	AP	External	It is recommended that you use thick analog power metal. When connected to PAD, the path should be kept as short as possible.
AVBB33A1	AG	External	Digital power and analog power are separately used.
AVSS33A1	AG	External	
AVSS33A2	DG	External	
AVDD33A2	DP	External	
AVDD12	DP	External	
EN	DI	Internal	EN, ENB are special function pin.
ENB	DI	Internal	Actually, EN = VDD (1.2V or 3.3V), ENB = VSS

HISTORY CARD

Version	Date	Modified Items	Comments
Ver1.0	02. 08. 30	Original version published (preliminary)	
Ver1.1	03. 04. 20	Feedback request is changed.	
Ver1.2	04. 02.18	Addition of test results and I/O pad information correction.	

NOTES